

 [Bookmark this page](#)

Upgrade

Let us now solve the companion system of the unforced coupled oscillator:

$$\frac{d}{dt}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \quad \text{where} \quad \mathbf{B} = \omega^2 \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The first step is to find the eigenvalues and eigenvectors of the companion matrix $\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix}$. We could work out the characteristic polynomial directly, but this involves computing the determinant of a 4×4 matrix. But in fact, because this is a companion matrix, its eigenvalues and eigenvectors can be written in terms of those of the smaller matrix $\mathbf{B}$ within it. To show this, let us inspect the following eigenvalue-eigenvector equation:

$$\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \lambda \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}.$$

To calculate the left hand side, we can use multiplication of block matrices, which works like usual matrix multiplication:

$$\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{y} \\ \mathbf{B}\mathbf{x} \end{pmatrix} \tag{4.83}$$

Hence, the eigenvalue-eigenvector equation becomes

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{B}\mathbf{x} \end{pmatrix} = \lambda \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \iff \begin{cases} \mathbf{y} &= \lambda\mathbf{x} \\ \mathbf{B}\mathbf{x} &= \lambda\mathbf{y}. \end{cases} \tag{4.84}$$

Now we perform a trick: we plug the first equation into the second to eliminate $\mathbf{y}$:

$$\mathbf{B}\mathbf{x} = \lambda^2\mathbf{x}.$$

This says that $\mathbf{x}$ is an eigenvector of $\mathbf{B}$ with eigenvalue λ^2 . Now we can solve for $\mathbf{y}$, namely $\mathbf{y} = \lambda\mathbf{x}$ and $\begin{pmatrix} \mathbf{x} \\ \lambda\mathbf{x} \end{pmatrix}$ is an eigenvector of $\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix}$.

The good news is that this procedure gives all the eigenvalues and eigenvectors of the 4×4 matrix.

If the matrix $\mathbf{B}$ has two eigenvalues and eigenvectors, let the eigenvalues be called λ_1^2, λ_2^2 , and the associated eigenvectors be $\mathbf{v}_1, \mathbf{v}_2$. As long as λ_1^2 and λ_2^2 are distinct and non-zero, the matrix $\begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{B} & \mathbf{0} \end{pmatrix}$ will have four eigenvalues:

$$\lambda_1, \quad -\lambda_1, \quad \lambda_2, \quad -\lambda_2 \tag{4.85}$$

with corresponding eigenvectors

$$\begin{pmatrix} \mathbf{v}_1 \\ \lambda_1\mathbf{v}_1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{v}_1 \\ -\lambda_1\mathbf{v}_1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{v}_2 \\ \lambda_2\mathbf{v}_2 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{v}_2 \\ -\lambda_2\mathbf{v}_2 \end{pmatrix}. \tag{4.86}$$

On the next page, we will carry out the computations and see that in this example, $\mathbf{B}$ does have distinct non-zero eigenvalues.

14. Solving the companion system of the coupled oscillator

Hide Discussion

Topic: Unit 3: Solving systems of first order ODEs using matrix methods / 14. Solving the companion system of the coupled oscillator

Add a Post

Show all posts ▼

by recent activity ▼

There are no posts in this topic yet.

✕

< Previous

Next >

